Joseph Catanzarite

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Summary

Dynamic Data Science Instructor with over 8 years of experience in curriculum development and teaching. Skilled in machine learning, deep learning, and predictive modeling, with a proven track record of enhancing learning outcomes. Successfully developed a Random Forest classifier achieving 97% accuracy for NASA Kepler mission data. With a unique blend of hands-on and theoretical knowledge, poised to contribute effectively to any educational or research institution's data science initiatives.

Work Experience

Data Science Instructor, Bloomtech, San Francisco

08/2021 – 12/2021

- Taught and developed Data Science and Machine Learning curriculum, supervised student projects, supervised work-study students.

Lead Data Science Instructor, Make School, San Francisco

11/2020 – 08/2021

- Developed and implemented comprehensive curriculum for the Data Science concentration, including courses such as Deep Learning and Computer Vision, Machine Learning in Production, and Introduction to Data Science
- Facilitated Data Science Technical Seminar as a Capstone Project, enabling students to apply theoretical knowledge in practical scenarios

Volunteer Data Scientist, DataKind Medic Mobile Project, Brooklyn

11/2019 – 07/2020

- Enhanced predictive models to optimize maternal and child health outcomes through data-driven insights

Volunteer Study Group Lead, TWiML & AI, St. Louis

07/2019 – 04/2021

- Organized and orchestrated remote AI study groups, leading an in-depth curriculum including Hands-On Machine Learning, A Code-first Introduction to Natural Language Processing (Fastai), and Deep Learning from the Foundations (Fastai)

Fastai International Fellow, University of San Francisco, San Francisco

10/2018 – 10/2019

Data Scientist, SETI Institute, Mountain View

01/2013 – 08/2017

- Contributed to the Kepler Science Team as part of NASA's Kepler Mission, building a Random Forest classifier for exoplanet candidates that achieved 97% accuracy

- Authored the report 'Autovetter Planet Candidate Catalog for Q1-Q17 Data Release 24' for NASA HQ
- Collaborated with the Modeling and Analysis team on Synthetic Flux-level Transit Injection Experiments, devising algorithms to estimate exoplanet detection efficiency based on signal-to-noise ratios for Kepler target stars
- Pioneered innovative applications of k-Nearest Neighbors (kNN), resulting in an 85% reduction in time to perform key experiments on the NASA Pleiades supercomputer
- Engineered and documented software algorithms for Kepler's Data Processing Pipeline in MATLAB under SVN (Subversion) version control, including Ghost Diagnostic Test and Ephemeris-Matching code as part of the Data Validation (DV) module, and developed the Thruster Firing Events Detector for the Photometric Analysis (PA) module, as well as a Wavelet-Based Denoiser for the Presearch Data Conditioning (PDC) module

Project Staff Scientist, Jet Propulsion Laboratory (JPL), Pasadena

01/2003 – 01/2013

- Led JPL's Analysis Team for NASA's Double Blind Test in Phases 1 and 2, coordinating across five teams to achieve consensus solutions
- Invented a novel algorithm for hierarchical detection and characterization of exoplanets based on joint astrometric and radial velocity data, playing a key role in proving SIM's capacity to detect Earth-mass planets
- Involved in the Advanced Mirror Development (AMD) Testbed Data Analysis Team, devising algorithms for wavefront and line-of-sight jitter characterization and demonstrating compliance with customer requirements through rigorous analyses
- Executed mission modeling efforts that assessed the exoplanet finding capabilities of astrometric, radial velocity, and coronagraphic missions, while also modeling the effects of starspots on exoplanet detection

Senior Engineer and Senior Application Developer, California Institute of Technology, Pasadena

01/1999 – 01/2003

- Conducted trade studies to analyze astrometric performance in relation to spacecraft instrument parameters
- Engineered data analysis algorithms for SIM's Micro Arc-second Metrology test bed
- Preprocessed WIRE (Wide Field Infrared Explorer) star-tracker data to support asteroseismology research

Professor of Physics and Astronomy, Cypress Community College, Cypress

09/1985 – 01/1999

- Chaired the Physical Sciences Department, overseeing curriculum development
- Educated students by teaching and developing comprehensive courses in Physics and Astronomy, both lecture and laboratory formats

Education

Master of Science, Electrical Engineering, University of Southern California,

Master of Science, Physics, University of Arizona,

Bachelor of Science, Mathematics, California Institute of Technology,

Skills

Machine Learning | Neural Networks | Computer Vision | Natural Language Processing | Time Series Analysis | Signal Processing | Bayesian Modeling | Statistics | Software Engineering | MATLAB | Python | SQL | TensorFlow | Communication | Curiosity | Continuous Learning | Exploration | Interpretation | Modeling | Mathematical Methods | Statistical Methods | Technical and Non-technical Audience Communication | Curriculum Development | Teaching | Leadership

Languages

English

Certificates

Practical Deep Learning for Coders 1 & 2, Fast.ai, (University of San Francisco)

2018

Machine Learning with TensorFlow on Google Cloud Platform Specialization, Coursera (Google)

2018

Deep Learning Specialization, Coursera (Andrew Ng, deeplearning.ai)

2018

SQL for Data Science, Coursera (UC Davis)

2018

A Crash Course in Causality: Inferring Causal Effects from Observational Data, Coursera (Univ. of Penn.)

2018

Improving Your Statistical Inferences, Coursera (Eindhoven University of Technology)

2018

Machine Learning Specialization, Coursera (University of Washington)

2017

Introduction to Computer Science and Programming Using Python, edX (MIT)

2016

Statistical Learning, Stanford University

2015

Introduction to Probability – The Science of Uncertainty, edX (MIT)

2015

Convex Optimization, Stanford University

2014

Discrete Time Signals and Systems, edX (Rice University)

2014

Digital Signal Processing, Coursera (Ecole Polytechnique Federale de Lausanne)

2013

Linear and Integer Programming, Coursera (University of Colorado at Boulder)

2013

Learning from Data, California Institute of Technology

2012